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Youth Players Basic Need Satisfaction, Enjoyment and Pleasure within daily team sport training sessions, and the relationships with Physical Load Engagement: An intensive within-person longitudinal study

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Abstract

Youth team-sport training sessions provide repeated opportunities for skill development, yet little is known about how athletes' momentary psychological experiences relate to their physical engagement in training. Fifty adolescent female handball players (M = 14.57, SD = 1.03) completed brief questionnaires before and after 11–12 consecutive team training sessions, while physical load (PlayerLoad™) and high-intensity events (> 1.5 m/s²) were recorded via inertial measurement units during training. Bayesian within-person analysis revealed temporal stability in basic need satisfaction, sport enjoyment, and feelings of pleasure before and after training. Before and after training, basic needs satisfaction was associated with sport enjoyment and pleasure. Pleasure before training was associated with physical load during training, which was further associated with pleasure after training. This highlights that youth athletes' basic need satisfaction and affective experiences are intertwined with their physical engagement in training, underscoring the importance of creating need-supportive training environments that foster both positive affect and meaningful behavioral involvement for athlete development in youth sport.

Introduction

Youth sport team training sessions offer repeated opportunities for athletes to develop skills through physical engagement with structured practice tasks. However, the literature on expertise and athlete development presents partly complementary and partly contrasting perspectives on the role of psychological experiences during training (Baker & Young, 2014; Baker et al., 2024; Côté et al., 2014; Ericsson et al., 1993).

Within the deliberate practice framework (Baker & Young, 2014; Ericsson et al., 1993), high performance is understood as the result of sustained, structured, and effortful practice accumulated over many years. From an expertise development perspective, repeated physical exposure to training tasks represents a fundamental behavioral pathway for skill acquisition (Ericsson et al., 1993; Baker & Young, 2014). Modern athlete monitoring technologies provide an opportunity to quantify physical engagement directly through measures of external physical load, such as accelerometer-derived PlayerLoad™ and counts of high-intensity actions (Boyd et al., 2011; Bourdon et al., 2017). Although such metrics are widely used in applied sport science to evaluate training and competition demands, their integration into motivational and developmental research remains limited (Baker & Young, 2014; Harwood et al., 2015). However, the deliberate practice framework places emphasis on the quantity, structure, and purposefulness of practice activities. Thus, motivation is conceptualized primarily as the performer's long-term commitment to improving performance. Accordingly, Baker and Young (2014) highlight that deliberate practice requires performers to maintain motivation over prolonged periods and to engage in practice specifically for the purpose of performance enhancement. In this view, enjoyment, pleasure, and social experiences are not central drivers of practice engagement. Rather, they are considered peripheral or even potentially distracting if they shift attention away from practice goals (Baker & Young, 2014). Thus, the psychological dimension emphasized in the deliberate practice literature concerns goal-directed

persistence over time, rather than the moment-to-moment quality of affect or perceived meaning during individual training sessions.

In contrast, ecological and developmental perspectives on youth sport (Côté et al., 2014) highlight the quality of athletes' participation experiences. These frameworks argue that learning opportunities arise when athletes actively interact with varied, meaningful, and challenging practice tasks. Ecological-developmental models similarly acknowledge the importance of active motor engagement, but they emphasize that the psychological meaning athletes assign to training tasks may influence how they engage with them (Côté et al., 2014). From this standpoint, the athlete's subjective experience of training such as whether the session feels meaningful, competence-enhancing or socially supportive shapes how they engage with practice activities. Rather than solely focusing on long-term commitment to improving performance, this perspective emphasizes that the psychosocial climate and experiential quality of individual training sessions influence how athletes behave and learn.

Although these contemporary frameworks of expertise and athlete development do not frame their arguments through self-determination theory (SDT) explicitly (Baker & Young, 2014; Baker et al., 2024; Côté et al., 2014), their emphasis on meaningfulness, perceived competence, and supportive social environments aligns closely with the concepts of basic psychological needs satisfaction (Deci & Ryan, 2000; Ryan & Deci, 2017). According to SDT, the satisfaction of the basic psychological needs for autonomy, competence, and relatedness energizes and sustains autonomous motivation, which can manifest emotionally through feelings of pleasure and enjoyment, and behaviorally through increased persistence and physical engagement (Fenton et al., 2016; Kalajas-Tilga et al, 2020; Ryan & Deci, 2017; Teixeira et al., 2012, 2021). Indeed, Quested et al. (2013a) demonstrated the importance of this distinction in their diary study of dancers' daily fluctuations in basic psychological need satisfaction related to their positive and negative affect across three distinct contexts (classes, rehearsals, and performances). The findings showed that higher satisfaction of the three basic psychological needs was consistently associated with greater positive affect and lower negative affect across all contexts (Quested et al., 2013a). However, the strength of these relationships differed depending on the situation. Competence satisfaction was most strongly linked to positive affect during performances, where demonstrating skill and effectiveness was central. Autonomy satisfaction was most influential in classes, where learning tasks allowed more personal choice and exploration. Lastly, relatedness need satisfaction showed the strongest association with positive affect during rehearsals, when cooperation and social connection with others were most salient.

To the best of our knowledge, however, only few studies have investigated the relation between autonomous motivation and objectively measured physical engagement in sports (Fenton et al., 2016; Harwood et al., 2015). Those studies found that perceived autonomy support from coaches in youth sport is positively and significantly related to athletes' autonomous motivation, which, in turn, is positively and significantly related to objective physical activity measured by accelerometers using the Actigraph GT3X (Fenton et al., 2016). Similarly, research in physical education shows that perceived autonomy support from teachers, parents, and peers is associated with higher physical activity, greater

intrinsic motivation, and stronger satisfaction of the needs for autonomy, competence, and relatedness (Kalajas-Tilga et al., 2020; Standage et al., 2012; Wang, 2017). Additionally, diary research in physical education with accelerometers shows that children's positive affect is followed by higher physical activity later the same day, which suggests that pleasant feelings can precede and accompany increased physical activity engagement (Dunton et al., 2013). Similar patterns have also been observed among young athletes across different sports. For instance, studies have found that higher levels of sport enjoyment are positively associated with self-reported behavioral engagement and stronger intentions to continue participating (Hodge et al., 2023). However, the necessity to investigate and distinguish between feelings of pleasure and sport enjoyment becomes particularly important when studying affective experiences in different sport settings, as opposed to general exercise, health-promotion, or physical education contexts (Ekkekakis, 2009; Oliveira et al., 2015; Vallerand, 1997).

Situational pleasure/displeasure reflects immediate affective valence (Hardy & Rejeski, 1989), whereas sport enjoyment represents a more enduring cognitive–affective evaluation of sport participation as rewarding and worthwhile (Bugten et al., 2025a, 2025c; Scanlan & Simons, 1992; Scanlan et al., 2016). The Dual-Mode Theory of affective responses to exercise stipulates that affective valence becomes less pleasant above the lactate threshold and more pleasant below it, linking physiological intensity directly to feelings of pleasure and displeasure (Ekkekakis, 2009; Oliveira et al., 2015).

Recent research examining sport enjoyment alongside objective measures of physical engagement suggests that this relationship may not always be held in competitive sport contexts. For instance, in soccer, players rated small-sided games as more enjoyable than interval running, despite both forms of training producing similar physiological demands (Selmi et al., 2020). Likewise, in female basketball, small-sided games and high-intensity training elicited comparable heart rate responses and perceived exertion, yet the small-sided games were experienced as more enjoyable (Zeng et al., 2023). These findings indicate that when athletes enjoy their training and experience positive emotions, they can sustain equal or even greater levels of physical engagement. This highlights the importance of context, situation, and task structure in shaping how athletes interpret and respond to physical intensity. To the extent physical activity engagement above or below the lactate threshold feels pleasurable or unpleasurable may also depend on the sport-specific meaning of the activity and the athlete's satisfaction of competence, relatedness, and autonomy within it (Quested et al., 2013a, 2013b). In sport-specific contexts and settings, the momentary feelings of pleasure from basic psychological need satisfaction and sport enjoyment may override momentary feelings of physical discomfort, allowing athletes to continue exerting higher physical engagement even under physiologically demanding conditions.

Taken together, previous research highlights two important processes in athlete development. One is the physical engagement with practice tasks, and the second is the psychological experiences that accompany such engagement (Baker & Young, 2014; Baker et al., 2024; Côté et al., 2014). However, these contemporary frameworks of expertise and athlete development remain largely disconnected. The current study sits at the intersection of these perspectives, using SDT as a basis for understanding how

athletes' psychological experiences may relate to their behavioral engagement in training (Deci & Ryan, 2000; Ryan & Deci, 2017). Nonetheless, the extent to which basic psychological needs satisfaction and affective experiences such as enjoyment and pleasure correspond with session-specific variability in external physical load remains largely unknown, particularly in youth team-sport settings where training demands and players' psychological states fluctuate from session to session. Addressing this gap offers an opportunity to advance understanding of athlete development in how psychological experiences and physical engagement operate together in the context of youth sport team training sessions.

Purpose of Study

The present study has three primary objectives. The first objective is to examine the associations between youth players' basic psychological need satisfaction, sport enjoyment, and feelings of pleasure before and after team training. The second objective is to examine the associations between youth players' basic psychological need satisfaction, sport enjoyment, and feelings of pleasure before team training with physical load and intensity engagement during team training sessions. Finally, the third objective is to examine the associations between youth players' physical load and intensity engagement during team training with youth players' basic psychological need satisfaction, sport enjoyment, and feelings of pleasure after training sessions.

Method

Participants

The study involved female adolescent players drawn from two Norwegian handball teams competing in the Under-14 and Under-16 categories. Altogether, 50 players participated, with ages ranging between 14 and 16 years (M = 14.57, SD = 1.03). The players trained on a regular weekly schedule consisting of approximately three sessions of 90 minutes each and took part in local grassroot-level competitions that did not include formal performance selection criteria. In line with existing classifications, they were categorized as trained developmental adolescent athletes (McKay et al., 2022). Focusing on this population was considered particularly important, as female adolescents are identified as a group with elevated vulnerability to reduced participation and dropout from organized sport (Bergeron et al., 2015; Emmonds et al., 2024; Moulds et al., 2024).

Note that the same participant sample has been used in two other manuscripts by the authors (Bugten et al., 2025b; Bugten et al., 2025c). This information is reported to avoid potential dependency bias in future research that may treat these studies as independent. In the present study, however, only the data relevant to psychological experiences and objective physical engagement during training sessions were analyzed.

Design

The present study employed an intensive repeated-measures longitudinal design in which each of the 50 players contributed up to 11-12 training sessions over a four-week period. In the present study, most training sessions primarily consisted of structured technical drills (e.g., finishing, passing, positional exercises) with limited free gameplay toward the end of training. Within the sessions there were frequent stoppages and instructional breaks, with drills that often required players to wait their turn and substantial portions of time dedicated to coach explanations and demonstrations. Players generally took part in their usual roles and positions (e.g., defenders, midfielders, forwards), although on some occasions individuals temporarily covered other positions due to injuries or absences. The structure ensured that players were repeatedly exposed to comparable training formats, while still reflecting the natural variability of applied team sport practice. This design likely enhanced the validity of the physical load and intensity engagement indicators by reducing contextual variability (tactics, formation, role, drill design) and promoting more uniform participation across players (Akenhead & Nassis, 2016; Boyd et al., 2013; Luteberget & Spencer, 2017). Conversely, in less organized sessions emphasizing free play and tactical scenarios, PlayerLoad™ and counts of high-intensity events may provide a less direct proxy of individual physical engagement because movement demands are more strongly driven by tactical and positional context (Luteberget & Spencer, 2017).

Because the analyses were conducted at the within-person level, all stable individual characteristics (e.g., playing position, physical capacity, and typical workload patterns) are held constant statistically. Thus, the observed associations between basic need satisfaction, sport enjoyment, pleasure, and player load and intensity reflect intra-individual covariation and cannot be attributed to between-person differences such as position or role in the team. At the same time, we acknowledge that session-specific contextual factors, such as temporary changes in position or role during match play, may contribute to day-to-day variation in physical load and intensity. In our design, such variations are treated as part of the natural within-person variability we aimed to capture. Consequently, the results should be interpreted as indicators within the realities of normal youth team training sessions.

Procedure

Ethical clearance for the study was granted by the University of Agder Research Ethics Committee and the Norwegian Agency for Shared Services in Education and Research. Following approval, contact with potential participants was established in collaboration with the coaches of two Norwegian female adolescent handball teams. The coaches distributed information about the project to players and parents, after which 50 players agreed to take part. Both parental consent and player assent were obtained prior to participation. All participants were informed that their involvement was voluntary, that their responses would remain confidential, and that they could discontinue participation at any time without any negative consequences.

Data collection took place over a four-week period and included 11 to 12 consecutive training sessions for each team. Approximately 15 minutes before the start of each session, players completed a brief digital questionnaire assessing basic psychological need satisfaction (autonomy, competence, and

relatedness), sport enjoyment, and feelings of pleasure or displeasure. The same questionnaires were administered within 15 minutes after each training to capture post-training states. In cases where players came to training after the session had begun, the start of their participation was postponed until they had completed the pre-training assessment. All assessments were carried out at the training venue to minimize disruption, and players completed all self-report measures under the supervision of the research team.

During all training sessions, physical engagement was continuously monitored using inertial measurement units (IMUs; OptimEye S5 Firmware version 7.40, Catapult Sports, Australia), which each player wore in a lightweight, custom vest positioned between the shoulder blades beneath the team jersey. Each player was assigned a dedicated IMU device that they wore in every training session to ensure consistency in measurement across sessions. The IMUs recorded information from an accelerometer, gyroscope, and magnetometer, all sampling at 100 Hz, enabling precise quantification of acceleration, deceleration, rotational motion, and impact forces. Prior to data collection, players were accustomed to wearing the sensors during regular training sessions to minimize any potential influence on performance. After each session, all data were downloaded from the units and processed through Catapult Openfield software (version 2.5.0, Catapult Sports, 2025). Player substitutions and injuries were manually logged, and only on-field activity was retained for analysis. To further ensure data accuracy detailed notes were taken so that any uncertainties related to substitutions or injuries could be verified and corrected afterward. The research procedures were otherwise noninvasive and did not interfere with the normal flow of session play or preparation routines.

Measures

All questionnaires were translated into Norwegian and carefully adapted to the age group and sport-training context. Items were modified to reflect situational experiences during training sessions, in line with Vallerand's (1997) recommendation to distinguish between situational, contextual, and global motivational levels. For instance, players were asked to rate how they felt at that moment in training rather than how they generally felt about their sport. Reliability for all multi-item scales was evaluated using McDonald's omega (ω), computed from standardized parameter estimates in confirmatory factor analyses (CFA) based on responses from one training session during the first week (Geldhof et al., 2014; McDonald, 1970). An omega coefficient of .70 or above was considered acceptable (Lance et al., 2006).

Basic Psychological Need States in Sport

Players' basic psychological need satisfaction of autonomy, competence, and relatedness was assessed using nine items from the Psychological Need States in Sport Scale (PNSSS; Bhavsar et al., 2020). Items were selected for clarity, brevity, and their ability to represent the essential features of each psychological need (Fisher et al., 2016; Martela & Ryan, 2024; Song et al., 2022). Players indicated their agreement on a 7-point Likert scale (1 = *strongly disagree*, 7 = *strongly agree*). Example items include: "In the sport I'm currently playing, I feel free to make choices regarding how I train" (autonomy), "In the sport I'm currently playing, I am able to overcome challenges" (competence), and "In the sport I'm

currently playing, I feel supported" (relatedness). Reliability was acceptable for all three subscales: autonomy (ω = .76), competence (ω = .77), and relatedness (ω = .89).

Sport Enjoyment

Sport enjoyment was measured using a single, situationally adapted item from the Sport Commitment Questionnaire (SCQ; Scanlan et al., 2016): "Playing this sport makes me happy." Responses were recorded on a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*). As the measure contained a single item, internal consistency was not assessed (Allen et al., 2022; Fischer et al., 2016; Song et al., 2022). This item was chosen for its simplicity, clarity, and established suitability for use in repeated-measures designs: Prior research has demonstrated strong test–retest reliability and construct validity for this single-item indicator of sport enjoyment (Bugten et al., 2025b; Schuuman & Hamaker, 2019).

Feelings of Pleasure and Displeasure

Feelings of pleasure and displeasure were assessed as players' immediate affective states during training using the Feeling Scale (Hardy & Rejeski, 1989). The scale captures the core affective valence dimension, ranging from very negative (displeasure) to very positive (pleasure), reflecting how players feel in the present moment of activity. Players responded to the single item "How do you feel right now?" on an 11-point bipolar scale ranging from –5 (*very bad*) to +5 (*very good*), with 0 representing a neutral feeling state. The measure has demonstrated strong validity and sensitivity for detecting momentary affective responses in sport and exercise settings (Ekkekakis & Petruzzello, 1999; Hardy & Rejeski, 1989; Stevens et al., 2020). Consistent with its single-item format, internal reliability was not assessed; however, the Feeling Scale has shown robust test–retest reliability and responsiveness to changes in exercise intensity and motivational states in previous research (Ekkekakis et al., 2004; Stevens et al., 2020).

Physical Load and Intensity

After each session, data for PlayerLoad™, accelerations, changes of direction, and decelerations were downloaded from the OptimEye S5 devices and edited in the OpenField Software to adjust for actual playing time. Acceleration, deceleration and change of direction were computed into a summary variable of high intensity events (HIEs >1.5 m/s) and together with PlayerLoad™ used in the analysis. The PlayerLoad™ metric provides an accelerometer-based estimate of a player's external physical load, calculated as the square root of the summed, squared instantaneous rates of change in acceleration across the three movement planes (x, y, z), divided by 100 (Boyd et al., 2011, 2013). This composite value represents the overall mechanical stress experienced by the player. The detection of accelerations, changes of direction, and decelerations events relied on synchronized data from the accelerometer (for magnitude) and from the gyroscope and magnetometer (for directional information). Each event was expressed as a change in velocity (m/s) within the mediolateral and anteroposterior axes, and directionality was computed to classify the event as an acceleration, change of direction, or deceleration. All movement events exceeding 1.5 m/s were included in this analysis, and their total count represented

the number of HIEs. To account for differences in individual playing time, all metrics were normalized per minute of on-field activity.

The PlayerLoad™ per minute variable has been validated as a reliable indicator of exercise intensity in Australian football (Boyd et al., 2013) and is also supported as a reliable measure of physical performance monitoring in handball (Luteberget & Spencer, 2017; Luteberget et al, 2018). Validation data from Luteberget et al. (2018) confirmed strong interdevice reliability, with a coefficient of variation of 3.9% and an intraclass correlation coefficient (ICC) of .98 for HIEs (>2.5 m/s) and a coefficient of variation of 0.9% and ICC of .99 for PlayerLoad™ per minute. In the present study, high-intensity events were defined using a lower velocity threshold (>1.5 $m \cdot s^{-2}$) to better capture meaningful movement patterns in adolescent players. While Luteberget and Spencer (2017) employed a higher threshold of >2.5 $m \cdot s^{-2}$ in elite adult players, recent methodological reviews have emphasized the importance of adjusting such cut-points according to age, sex, and competitive level (Delves et al., 2023; Aandahl et al., 2025).

Data Analysis

Given the nested structure of the data, with repeated training sessions nested within players, Bayesian multilevel structural equation modeling (MSEM) was employed to examine both psychological and physical relationships. Analyses were performed in M*plus* (Version 8.3; Muthén & Muthén, 1998-2022) using a Bayesian estimator with diffuse, non-informative priors. Separate models were estimated for relatedness, competence, and autonomy to test the hypothesized associations among psychological and physical variables. As demonstrated in Figure 1, each model simultaneously included before-training variables (basic need satisfaction, pleasure, and enjoyment), during-training variables (PlayerLoad™ and high-intensity events), and after-training variables (basic need satisfaction, pleasure, and enjoyment). This analytic structure does not reveal any causal relations (Curran & Bauer, 2011; Stenling et al., 2017), but allows for testing the psychological pathways linking basic psychological needs, pleasure, and enjoyment across time (before and after training), as well as the pathways linking psychological variables with physical load and intensity engagement during training.

Figure 1

Overview of study variables measured before, during, and after team-sport training sessions.

Note. Solid boxes represent the observed variables at each measurement point. Dashed arrows illustrate the within-person associations tested in the study's models. PlayerLoad™ represents the accelerometer-derived index of external physical load, and high-intensity events reflect accelerations exceeding 1.5 m/s^2.

Posterior predictive checking (PP*p*) was used to evaluate model fit, with values near .50 indicating good fit. Additional model adequacy indices included the Deviance Information Criterion (DIC) and the estimated number of parameters (pD). Parameters were considered credible when the 95% credible

interval (CI) did not include zero. Standardized coefficients (STDYX) are reported for interpretability. Analyses focused primarily on within-person effects to capture session-to-session fluctuations in players' motivation, affect, and physical workload, while stable between-person differences were modeled using random intercepts. Descriptive statistics and correlations were examined prior to modeling to ensure data quality, confirm normality, and verify sufficient within-person variability for reliable estimation.

Results

Table 1 presents the descriptive statistics and intraclass correlation coefficients (ICCs) for all study variables across training sessions.

Note. Values are presented as means (M) and standard deviations (SD). Intraclass correlation coefficients (ICC) reflect within-person stability across training sessions. PlayerLoad™ is expressed in arbitrary units (AU) per minute and represents an accelerometer-derived composite index of external physical workload. High-intensity events represent the number of accelerations exceeding 1.5 m/s² per minute of training.

The ICCs indicated that between 8% and 69% of the total variance was attributable to between-person differences, depending on the variable. Specifically, ICCs were lowest for high-intensity events (ICC = .08) and feelings of pleasure (ICC = .29), suggesting greater within-person variability across sessions, while ICCs for basic psychological need satisfaction ranged from .62 to .69, indicating more than stable individual differences.

Bayesian Multilevel Analysis

All Bayesian multilevel models demonstrated excellent convergence and acceptable fit. Posterior predictive *p*-values were close to .50 (Relatedness: PP*p* = .500; Competence: PP*p* = .495; Autonomy: PP*p* = .500), and the 95% CIs for the difference between observed and replicated chi-square values were narrow, indicating strong model fit.

Associations Between Basic Need Satisfaction, Feelings of Pleasure, and Sport Enjoyment Before and After Team Sport Training Sessions

The analyses revealed strong temporal stability across all psychological constructs. As demonstrated in Table 2, basic psychological need satisfaction before training significantly predicted corresponding need satisfaction after training (Relatedness: β = .68, 95% CI [.56, .74]; Competence: β = .63, 95% CI [.56, .74]; Autonomy: β = .65, 95% CI [.56, .74]). Similarly, pleasure and sport enjoyment demonstrated substantial within-person stability across sessions (pleasure: β = .55, 95% CI [.47, .63]; enjoyment: βs = .45–.49, 95% CI [.36, .56]).

INSERT TABLE 2 HERE

Pre-training pleasure was consistently associated with positive affective outcomes. Players reporting greater pleasure before training experienced higher post-training pleasure (β = .55, 95% CI [.47, .63]) and enjoyment (βs = .13–.15, 95% CI [.04, .24]). Likewise, greater pre-training enjoyment were associated with higher post-training pleasure in the relatedness and competence models (βs = .09–.10, 95% CI [.01, .17]). As shown in Table 3, each basic need also demonstrated unique associations with affective states. Higher pre-training competence satisfaction was associated with greater post-training enjoyment (β = .15, 95% CI [.07, .24]), and greater autonomy before training was associated with greater post-training pleasure (β = .09, 95% CI [.01, .17]). Post-training need satisfaction correlated positively with concurrent pleasure and enjoyment across all needs (Relatedness–Pleasure: β = .15, 95% CI [.06, .24]; Competence–Pleasure: β = .23, 95% CI [.13, .31]; Autonomy–Pleasure: β = .11, 95% CI [.01, .20]; Competence–Enjoyment: β = .08, 95% CI [.01, .19]; Autonomy–Enjoyment: β = .14, 95% CI [.04, .24]).

INSERT TABLE 3 HERE

Within the same measurement occasion, post-training pleasure and enjoyment were correlated (r = .24, 95% CI [.13, .33]). Similarly, pleasure and enjoyment before training were positively related (r = .33, 95% CI [.23, .42]). Furthermore, higher pre-training need satisfaction was associated with more positive affective states before training: relatedness with pleasure (r = .43, 95% CI [.34, .50]), competence with pleasure (r = .37, 95% CI [.28, .45]), and autonomy with pleasure (r = .36, 95% CI [.28, .44]).

Overall, these findings indicate that players who felt more competent, autonomous, and related before training also reported greater pleasure and enjoyment, and that these affective states remained stable and mutually reinforcing throughout the training sessions.

Associations Between Basic Psychological Need Satisfaction, Feelings of Pleasure, and Sport Enjoyment Before and After Training and Physical Engagement During Team Sport Training Sessions

In examining the psychological and physical associations, players' affective states before training predicted physical engagement during training. As demonstrated in Table 2, pre-training pleasure was positively associated with physical workload (PlayerLoad™) during training across all models (βs = .14–.17, 95% CI [.03, .28]). In turn, greater workload was associated with higher post-training pleasure (β = .09, 95% CI [.01, .17]), suggesting a reciprocal relation between positive affect and physical engagement. These associations indicate that players who experience more pleasure prior to training tended to engage more physically during training, and that this physical engagement subsequently reinforced positive affect after training. However, no significant effects were observed between high-intensity events (> 1.5 m/s) and any psychological variables.

Discussion

The aim of this study was to examine associations of youth handball players' basic psychological needs satisfaction, feelings of pleasure, and sport enjoyment before and after training, and how these psychological experiences are associated with physical load and intensity engagement during team-

sport training sessions. Before and after training, players' basic needs satisfaction was associated with sport enjoyment and pleasure. Additionally, satisfaction of players' basic psychological needs, sport enjoyment, and pleasure before training was associated with basic needs satisfaction, sport enjoyment and pleasure after training. Furthermore, feelings of pleasure before training was associated with physical load during training. Physical load was in turn associated with pleasure after training. Together, these findings offer new insights into how young players' psychological experiences before and after training are associated with their physical engagement during team-sport training sessions. They highlight two important processes in youth sport development, the perceived psychological quality of training experience and the behavioral exposure through which skill development occurs (Baker & Young, 2014; Baker et al., 2024; Côté et al., 2014).

The first objective in the current study concerned the associations between psychological variables before and after training. Associations from before to after training in basic need satisfaction, pleasure, and enjoyment suggest that players' initial states when entering training sessions play a substantial role in shaping how they experience the session overall. This aligns with SDT (Ryan & Deci, 2017) and Vallerand's (1997) hierarchical model, which emphasize that momentary experiences are embedded in broader motivational patterns, but can still fluctuate meaningfully at the situational level. The stability observed in the present study implies that the first moments of a training session, such as how players are welcomed, how training tasks are introduced, and the immediate sense of autonomy, relatedness, and competence players feel, may set the tone for the entire session. This pattern resonates with previous diary research on youth sport athletes showing that daily fluctuations in basic psychological need satisfaction are associated with corresponding changes in positive affect and enjoyment (Bugten et al., 2025c; Quested et al., 2013a).

The findings also demonstrate that the three basic needs do not play identical roles across time in youth team training sessions. Competence need satisfaction before training was associated with greater enjoyment after training, while autonomy need satisfaction before training was associated with higher pleasure after training. These differential patterns highlight that each need may be more closely linked to certain affective outcomes under specific situational conditions. The team training sessions studied in the current study was mainly drill-oriented, where often technical mastery and progress were emphasized. In these specific team training settings feeling competent appears especially important for shaping how enjoyable the session feels. This aligns with past evidence that competence is the most immediate driver of sport enjoyment in structured training environments (Bugten et al., 2025c; Quested et al., 2013a). Autonomy, on the other hand, was more closely tied to momentary pleasure, supporting the idea that having some sense of choice or ownership within team training contributes to feeling pleasure during and after the activity (Ryan & Deci, 2017). Relatedness need satisfaction coming into training was not associated with neither sport enjoyment or feelings of pleasure after training. However, relatedness need satisfaction after training was positively associated with concurrent feelings of pleasure and enjoyment after training, underscoring the importance of social connectedness for maintaining a positive affective experience after training. The importance of social bonding and

relatedness support after training sessions aligns with previous diary study findings in youth sport (Bugten et al., 2025a).

A further contribution of this study lies in distinguishing between pleasure and enjoyment as two related but distinct affective states. Pleasure, measured as immediate core affect, showed stronger within-person stability and association of pleasure after the training session. Enjoyment, reflecting a more evaluative sense that participation was worthwhile and rewarding, was somewhat less stable and more selectively related to specific variables such as need for competence. These distinctions align with previous theoretical proposals that pleasure represents a direct emotional response to the current activity, whereas enjoyment integrates cognitive appraisals of meaning and satisfaction (Scanlan & Simons, 1992; Ekkekakis, 2009). By measuring both constructs before and after team training sessions, the present study provides situational evidence that these two affective experiences co-exist and influence each other in complementary ways.

The second and third objective concerned whether players' psychological experiences were associated with their physical engagement during team training, and, in turn, whether players' physical engagement during training was associated with psychological experiences after training. Our findings demonstrated that players' feelings of pleasure before training were tied to their physical engagement during training. Specifically, pleasure before training was associated with physical load during training. In turn, physical load during training was associated with feelings of pleasure after training. This is consistent with SDT's proposition that positive affect emerges from need satisfaction and functions as an energizing factor that facilitates behavioral engagement (Ryan & Deci, 2017; Hodge et al., 2023). It also aligns with ecological-developmental perspectives suggesting that positive psychological experiences may support athletes' willingness to invest effort and engage actively with training tasks (Côté et al., 2014).

The reciprocal pattern may also suggest that affective and physical engagement operate as a reinforcing cycle: when players start training in a pleasant emotional state, they are more likely to invest energy and engagement, and the resulting exertion contributes to renewed positive affect afterward. These findings extend previous studies that found associations between youth athletes intrinsic motivation, positive affect and increased physical activity levels (Fenton et al., 2016) by demonstrating this process within youth team-sport training sessions. This highlights the importance of considering affect as a dynamic input, not merely an outcome, of physical engagement in youth sport. Additionally, our findings challenge previous assumptions rooted in dual-mode theory, that feelings of pleasure are determined primarily by whether physical demands exceed the lactate threshold (Ekkekakis, 2009). Indeed, the reciprocal pattern highlights that this relationship may not always be held in youth sport context within team training sessions, where psychological experiences contribute to shaping how players interpret and respond to physical load. Consistent with prior studies (Quested et al., 2013a; Quested et al., 2013b), the extent to which physical engagement feels pleasant or unpleasant appears to also depend on the sport-specific meaning of the activity and the athlete's satisfaction of competence, relatedness, and autonomy. In the current study's youth sport context, the momentary feelings of pleasure from basic psychological need satisfaction and sport enjoyment seem to override momentary feelings of physical discomfort, making it

more rewarding (or less costly) for athletes to continue exerting higher physical load engagement in team training sessions.

Lastly, a central contribution of this study is its attempt to integrate perspectives that have remained largely separate in the youth sport athlete development literature. Deliberate practice research emphasizes the accumulation of structured, effortful physical engagement as a key route toward expertise (Ericsson et al., 1993; Baker & Young, 2014). In contrast, ecological-developmental models highlight the importance of athletes' experiential interpretation of training tasks such as their enjoyment, sense of meaning, and social connection (Côté et al., 2014). SDT complements these views by providing a theoretical account of how psychological need satisfaction shapes affective experiences and behavioral engagement in youth sport (Ryan & Deci, 2017). The findings of the current study provide empirical support for this integration by demonstrating how athletes' affective experiences and basic psychological need satisfaction are associated to their physical engagement during team training and to the stability of these experiences across sessions. Together, these results suggest that physical workload, affective states, and need satisfaction should not be viewed as separate developmental processes, but as dynamically intertwined components of youth sport participation and development.

Practical Implications

The findings of this study hold several meaningful implications for how coaches, sport psychologists, and practitioners can create and facilitate youth team-sport training sessions that support both athletes' well-being and sport engagement. The results suggest that psychological experiences at the start of a training session are particularly influential, shaping not only how athletes feel during and after training but also how much physical engagement they invest during training.

One implication concerns the differentiated role of the three basic psychological needs. The results showed that competence need satisfaction before training was associated with sport enjoyment after training, whereas autonomy need satisfaction before training was associated with pleasure after training. This suggests that coaches can deliberately target specific needs depending on their goals for the session. If the objective is to increase athletes' momentary sport enjoyment of how rewarding or satisfying the session feels then training should be structured to highlight progress and mastery (Scanlan et al., 2016; Ryan & Deci, 2017). This can be achieved by providing clear performance feedback, setting achievable but challenging goals, and designing drills where athletes can see tangible improvement (Bhavsar et al., 2019). Conversely, if the goal is to enhance the pleasantness of the session and sustain positive mood states, providing opportunities for choice, creativity, and self-expression may be most effective (Bhavsar et al., 2019).

Second, the reciprocal association between pleasure before training, physical load during training, and pleasure after training suggests that well-designed training tasks that support basic psychological needs and promote active involvement may reinforce positive affective experiences and potentially foster longer-term engagement in youth sport. Beyond the established health benefits of meeting recommended levels of physical activity in youth sport participation (Fenton et al., 2016; Moulds et al.,

2024), our findings indicate that greater physical load engagement during team training can itself be associated with enhanced momentary feelings of pleasure. This highlights the value of structuring youth team training sessions to facilitate sustained physical involvement, for example, by minimizing unnecessary interruptions or prolonged coach talk, and maximizing athletes' opportunities to interact with practice tasks. Such conditions may help optimize both the experiential quality of training and the behavioral exposure through which skill development occurs (Baker & Young, 2014; Baker et al., 2024; Côté et al., 2014).

Limitations and Future Directions

Although the present study provides new insight into how psychological experiences and physical engagement interact within youth team-sport training sessions, several limitations should be acknowledged. First, the design was observational, meaning that the temporal patterns identified such as the link between pre-training pleasure, physical load, and post-training affect, cannot be interpreted as causal (Curran & Bauer, 2011; Stenling et al., 2017). Future studies should use experimental or micro-intervention designs to manipulate pre-training affect or need-supportive coaching behaviors and directly test their causal influence on engagement and workload.

Second, the indicators of physical load and intensity engagement used (PlayerLoad™ and high-intensity events) reflect only certain aspects of physical engagement and may be influenced by drill design, tactical context, and playing position (Luteberget & Spencer, 2017). Indeed, the team training sessions in the present study involved frequent stoppages and instructional breaks, with drills that often required players to wait their turn and substantial portions of time dedicated to coach explanations and demonstrations. Although the high-intensity threshold (>1.5 m/s) was adjusted as recommended for the participants' sex, age group, and competitive level (Delves et al., 2023; Aandahl et al., 2025), it might have been too low to meaningfully distinguish between intensity levels. This may indicate that high-intensity events are highly variable in this population and within the types of training activities conducted, making them less sensitive as indicators of players' behavioral engagement. It is also possible that players' motivational and affective experiences are more closely related to their overall physical load, rather than to short bursts of high-intensity movement performed in isolation. Therefore, future studies should investigate the potential modification of these metrics, integrating more contextual information or physiological data to create a more complete picture of physical engagement during training sessions.

Additionally, this study did not measure environmental factors such as coach autonomy support or peer interactions, which may also shape the satisfaction of basic needs and affective states (Bhavsar et al., 2019; Harwood et al., 2015; Ryan & Deci, 2017). Including observational or self-report measures of coaching climate and social dynamics would strengthen future models by clarifying the situational mechanisms driving session-to-session changes. Furthermore, the sample consisted of adolescent female handball players from Norwegian club settings, which limits generalizability. Replication in other

sports, competitive levels, and cultural contexts is needed to test whether the same within-session dynamics occur elsewhere.

Finally, although this study focused on immediate, situational processes, motivation and affect in sport also unfold over longer timescales (Vallerand, 1997). Future longitudinal designs could examine whether repeated experiences of need satisfaction and positive affect across training sessions accumulate into broader motivational patterns, athletic development, or sport retention over a season (Baker et al., 2014; Curran & Bauer, 2011; Côté et al., 2014; Stenling et al., 2017). By combining experimental, longitudinal, and mixed-methods approaches, future research can build on these findings to more fully explain how momentary psychological states translate into sustained physical engagement, well-being, and athlete development in youth sport.

Conclusive Remarks

The present findings showed that positive affective states (pleasure) before youth team training is associated with behavioral engagement (physical load) during training, and that physical load engagement during training in turn is associated with feelings of pleasure after training. These results suggest that the momentary experiential quality of training plays a role in shaping how much physical engagement young players invest in practice, thereby shaping their behavioral exposure to task activities through which skills may develop.

Declarations

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Data availability

The datasets generated and analyzed during the current study are not publicly available due to considerations of participant confidentiality and the lack of deposition in a public data repository. However, the data may be obtained from the corresponding author (J.B.B.) upon reasonable request and subject to the approval of the relevant institutional bodies.

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Tables

Tables 1 to 3 are available in the supplementary files section

Figures

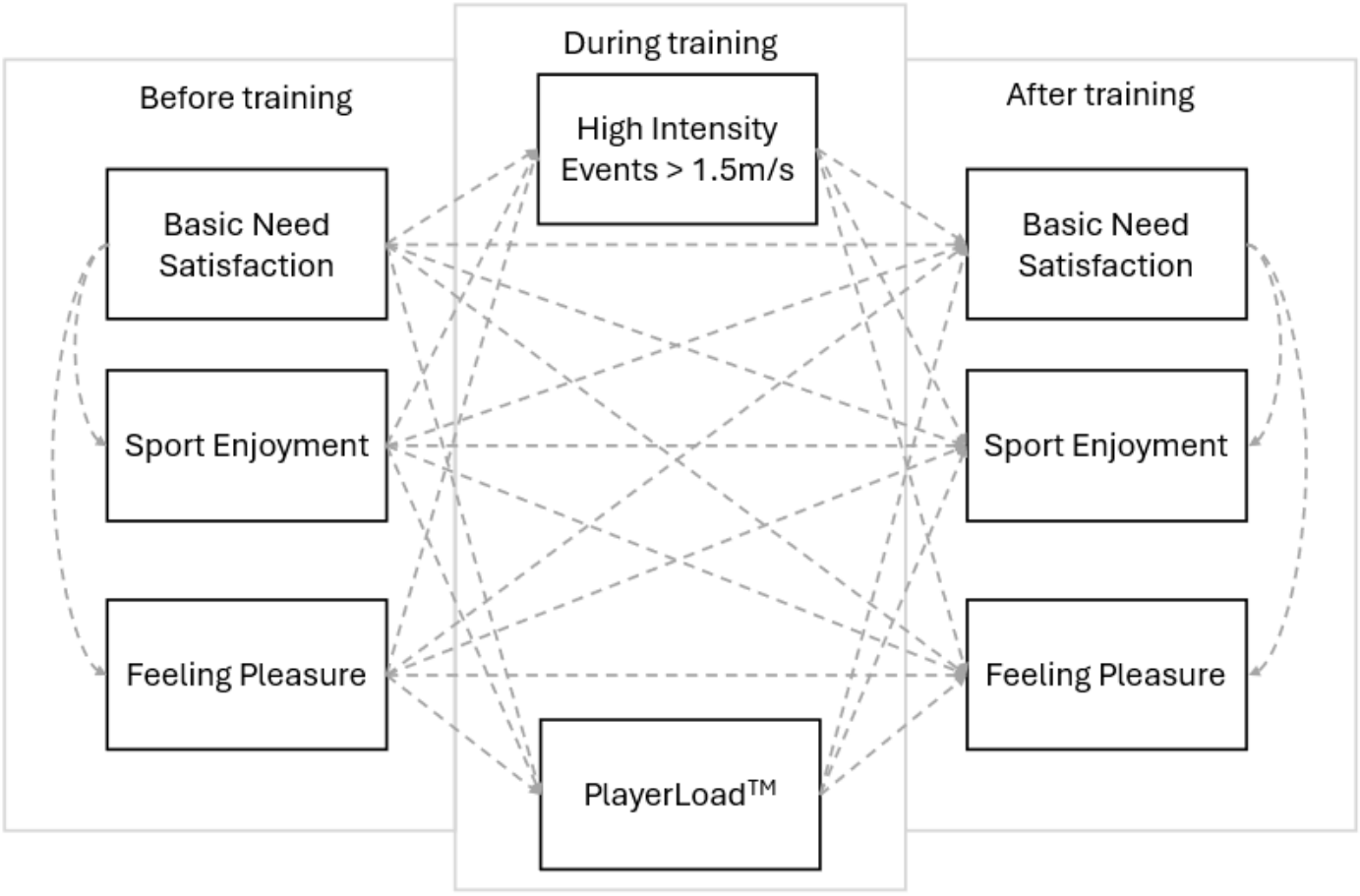


Figure 1

Overview of study variables measured before, during, and after team-sport training sessions.

Note. Solid boxes represent the observed variables at each measurement point. Dashed arrows illustrate the within-person associations tested in the study's models. PlayerLoad™ represents the accelerometer-derived index of external physical load, and high-intensity events reflect accelerations exceeding 1.5 m/s^2.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- Tables.docx
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